

Problem 1 MLE for Exponential distribution [30 marks]

In this problem you will consider the maximum likelihood estimate (MLE) of the parameters of a *Exponential Distribution*.

The exponential distribution is a distribution of the time interval between consecutive events occurring in a Poisson process. A Poisson process is a process where events occur continuously at some constant average rate. For example, if we model a telephone switch as a Poisson process, then the time between incoming calls can be modeled as an exponential distribution.

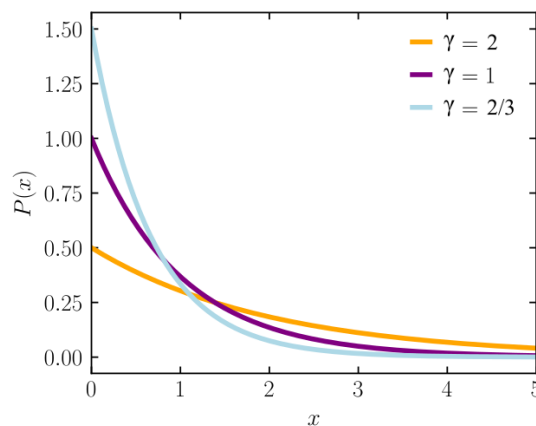
Formally, the exponential distribution is:

$$p(x|\gamma) = \begin{cases} \frac{1}{\gamma} e^{-x/\gamma}, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad (1)$$

where $\gamma > 0$ is the scale parameter or average interval length. The mean and variance of the exponential distribution are

$$E[x] = \gamma, \quad \text{var}(x) = \gamma^2. \quad (2)$$

Here is a plot of the exponential distribution for different values of γ :



Suppose we have a set of N samples of time intervals, $\mathcal{D} = \{x_1, \dots, x_N\}$ with $x_i \geq 0$.

- (a) [5 marks] Write down the log-likelihood of the data $\mathcal{D}$, i.e., $\log p(\mathcal{D}|\gamma)$.
- (b) [5 marks] Write down optimization problem for maximum-likelihood estimation of the parameter γ .
- (c) [15 marks] Derive the MLE for the parameter γ .
- (d) [5 marks] What is the intuitive interpretation of the derived MLE for γ .

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(a) [5 marks] The log-likelihood is

$$\log p(\mathcal{D}|\gamma) = \sum_{i=1}^N \log p(x_i|\gamma) \quad (1)$$

$$= \sum_{i=1}^N \log\left[\frac{1}{\gamma} e^{-x_i/\gamma}\right] \quad (2)$$

$$= \sum_{i=1}^N \left[-\log \gamma - \frac{x_i}{\gamma}\right] \quad (3)$$

$$= -N \log \gamma - \frac{1}{\gamma} \sum_{i=1}^N x_i \quad (4)$$

$$= -N \log \gamma - \frac{1}{\gamma} \bar{x}, \quad (5)$$

where we define $\bar{x} = \sum_{i=1}^N x_i$ for convenience.

(b) [5 marks] The MLE optimization problem is:

$$\hat{\gamma}_{MLE} = \underset{\gamma}{\operatorname{argmax}} \log p(\mathcal{D}|\gamma) \quad (6)$$

$$= \underset{\gamma}{\operatorname{argmax}} -N \log \gamma - \frac{1}{\gamma} \bar{x}. \quad (7)$$

(c) [15 marks] Taking the derivative of the objective and setting to zero,

$$\frac{\partial}{\partial \gamma} \left\{ -N \log \gamma - \frac{1}{\gamma} \bar{x} \right\} = \frac{-N}{\gamma} + \frac{1}{\gamma^2} \bar{x} = 0 \quad (8)$$

Multiplying both sides by γ^2 ,

$$-N\gamma + \bar{x} = 0 \quad (9)$$

$$N\gamma = \bar{x} \quad (10)$$

$$\Rightarrow \hat{\gamma}_{MLE} = \frac{\bar{x}}{N} = \frac{1}{N} \sum_{i=1}^N x_i \quad (11)$$

(d) [5 marks] The ML estimate $\hat{\gamma}_{MLE}$ is the sample mean of the population (the observed lengths). Thus MLE computes the sample mean of the data and fits the exponential distribution with the same mean.

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